

SecureSwipe:

Credit Card Fraud Detection System Using Machine Learning

Nishu

Mathematics and Computing

Indira Gandhi Delhi Technical University for Women

Delhi, India

nishu045btmac25@igdtuw.ac.in

Pratibha Gupta

Mathematics and Computing

Indira Gandhi Delhi Technical University for Women

Delhi, India

pratibha049btmac25@igdtuw.ac.in

Abstract

Credit card fraud has become a serious problem because many people have started using digital payment methods for their day to day transactions. This project uses machine learning techniques to identify fraudulent credit card transactions. We trained and tested our model using the European Credit Card Fraud dataset, which is available on kaggle containing 284,807 transactions and 31 features.

This project includes understanding data, preprocessing, exploratory data analysis, feature transformation, and testing of different machine learning models such as Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Gradient Boosting, and XGBoost.

The performance of each model was compared using Accuracy, Precision, Recall, F1-score, and ROC-AUC. Among all the models, Random Forest gave the best results. It resulted in an accuracy of 99.92%, precision of 96.5%, recall of 91.8%, F1-score

of 94.1%, and a ROC-AUC score of 0.996. These results show that the Random Forest model is very effective in detecting fraudulent credit card transactions, even when the dataset is highly imbalanced.

Keywords

Credit Card Fraud Detection,

Machine Learning,

Random Forest,

Class Imbalance,

Financial Security,

ROC-AUC,

XGBoost

I. Introduction

Digital payment systems have made it easier and faster for people to make payments online. They provide a secure and convenient way to complete transactions. However, as the number of online transactions has increased, cases of credit card fraud have also increased. These frauds can cause massive financial losses to banks, businesses, and customers. Because of this, fraud detection systems have become crucial for keeping online payment systems safe and trustworthy.

Credit card fraud detection is difficult because fraud transactions are very few compared to legitimate transactions. In this project, the dataset contains 284,807 transactions, but only 492 of them are fraudulent. This large difference creates a class imbalance, which makes it harder for

machine learning models to identify fraud correctly. If this issue is not taken into account properly, the model may classify most transactions as legitimate and fail to detect fraud.

Traditional systems are not very effective because fraud methods keep changing over time. Machine learning provides a better solution by learning patterns from transaction data and identifying suspicious activities more accurately.

In this project, different machine learning models are trained and compared using the same process. After comparing all the models, Random Forest gave the best results and was selected as the most suitable model for credit card fraud detection.

II. Research Contributions

Novelty of the Work:

The main contribution of our project is that it does more than just comparing different machine learning models. It follows a complete process that includes data preprocessing, feature analysis, training and testing different models, and comparing their performance using various evaluation metrics. It also explains which features are most important in detecting frauds and discusses the practical use of the model, its limitations, and possible improvements for future work. This makes our study more useful and complete than a simple model comparison.

III. Related Work

Credit card fraud detection has been widely studied because online transactions continue to increase. Early research mainly used machine learning models such as Logistic Regression and Decision Trees due to their simplicity. Later, ensemble methods like Random Forest and Gradient Boosting achieved better detection performance and reduced false positives.

A major challenge in fraud detection is the imbalance between genuine and fraudulent transactions. Therefore, researchers commonly evaluate models using Precision, Recall, F1-score, and ROC-AUC instead of relying only on accuracy.

Recent studies have also explored deep learning models, including LSTM and Autoencoders, which can identify complex transaction patterns. Although these models often improve detection, they require higher computational resources and longer training time. Current research also focuses on Explainable AI and Graph Neural Networks to improve model transparency and adapt to evolving fraud patterns.

Despite these advances, class imbalance and changing fraud techniques remain open challenges. This motivates the comparison of multiple machine learning models using a common preprocessing and evaluation framework.

LITERATURE REVIEW:

TABLE I

Table I describes the Literature Comparison

Literature Review Matrix: Credit Card Fraud Detection using Machine Learning

S.N o.	Referen ce	Proble m Addres sed	Methodology/T echnique	Datase t	Results/Perfo rmance	Limitatio ns	Resear ch Gap
1.	Bhattach aryya et al., 2011	Detect fraud in imbalan ced transac tions	Random Forest, Logistic Regression	Europe an Credit Card	High detection; low false alarms	Small dataset	Real-time detectio n
2.	Dal Pozzolo et al., 2015	Class imbalan ce	SMOTE + RF	Europe an CC	Improved recall/precisio n	Overfittin g	Better imbalan ce learning
3.	Dal Pozzolo	Changi ng fraud	Cost-sensitive learning	Europe an CC	Better fraud ID	Static costs	Adaptiv e costs

	et al., 2014	patterns					
4.	Jurgovsky et al., 2018	Sequential fraud	LSTM	Transaction sequences	Higher F1-score	High computation	Lightweight DL
5.	Carcillo et al., 2019	Real-time fraud	Streaming ML	Financial stream	Low latency	Complex deployment	Edge systems
6.	Fiore et al., 2019	Anomaly detection	Autoencoder	European CC	High accuracy	Parameter tuning	Hybrid models
7.	Bahnsen et al., 2016	Cost-sensitive fraud	Decision Trees	Bank data	Reduced loss	Cost estimation	Dynamic optimization
8.	Whitrow et al., 2009	Feature engineering	Aggregation features	Card data	Improved performance	Limited features	Auto feature extraction
9.	Phua et al., 2010	Fraud survey	Comparative review	Multiple	Technique comparison	No experiments	Benchmarks
10.	Srivastava et al., 2008	Behavior modeling	Hidden Markov Model	Card transactions	Good sequential detection	Slow training	Hybrid sequence models
11.	Randhawa et al., 2018	Fraud detection	XGBoost	European CC	High ROC-AUC	Feature engineering	Explainable boosting
12.	Ahmed et al., 2016	Fraud survey	Data mining review	Various	Compared methods	No implementation	Evaluation framework
13.	Chen et al., 2018	Fraud detection	Gradient Boosting	Financial	High precision	Complexity	Fast boosting
14.	Roy et al., 2018	Ensemble fraud detection	Voting classifier	Kaggle	Better F1	Complex ensemble	Efficient selection
15.	Aleskerv et al., 1997	Early fraud detection	Neural Network	Banking	Better than rules	Old architecture	Modern DL
16.	West & Bhattach	Intelligent	Review	Multiple	Research trends	Limited DL	Emerging AI

	arya et al., 2016	fraud survey					
17.	Ghosh & Reilly et al., 1994	Neural fraud detection	Neural Network	Bank	High recognition	Small data	Deep learning
18.	Lebichot et al., 2020	Graph fraud	Graph Neural Network	Transaction graph	Better relational detection	Graph complexity	Scalable GNN
19.	Mishra et al., 2021	Explainable fraud	XAI + RF	Credit card	Interpretable	Lower accuracy	Explainable DL
20.	Kumar et al., 2022	Hybrid detection	CNN+LSTM	Kaggle	Improved accuracy	Training time	Lightweight hybrid

IV. Research Gap

Although many studies have improved credit card fraud detection, several challenges still remain. The main issue is class imbalance, as fraudulent transactions are much fewer than legitimate ones, it makes fraudulent ones difficult to detect. Some techniques improve detection but may increase model complexity.

Reducing false positives is another challenge, as incorrect fraud alarms can affect both customers and banks. Advanced models like ensemble methods provide better accuracy but require more computing power and training time.

Most existing studies also use different preprocessing methods and evaluation metrics, making fair comparison difficult. Therefore, a common framework is needed to compare machine learning models using the same evaluation methods to identify the most suitable model for credit card fraud detection.

V. Objectives of the Study

- To analyze the credit card transaction dataset and understand its characteristics.
- To preprocess the data and perform exploratory data analysis (EDA).
- To train machine learning models, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Gradient Boosting, and XGBoost.
- To evaluate and compare the models using Accuracy, Precision, Recall, F1-score, and ROC-AUC.
- To identify the most effective model for detecting fraudulent transactions.
- To develop a robust machine learning framework for credit card fraud detection.

VI. Materials and Methods

A. Dataset Description

The experiments were conducted using the European Credit Card Fraud Detection

dataset available on Kaggle. It contains 284,807 transactions, of which 492 are fraudulent and 284,315 are legitimate. Since fraudulent transactions represent only a small fraction of the data, the dataset is highly imbalanced.

For privacy reasons, most transaction features were transformed using Principal Component Analysis (PCA). Therefore, the dataset contains 28 anonymized features (V1–V28), along with Time and Amount. The target variable, Class, identifies legitimate transactions as 0 and fraudulent transactions as 1.

Because of the imbalance in the dataset, accuracy alone does not provide a complete picture of model performance. For this reason, Accuracy, Precision, Recall, F1-score, and ROC-AUC were used to evaluate the models.

The dataset represents real credit card transactions and provides a suitable basis for training and comparing different machine learning models under the same conditions.

B. Data Preprocessing

The dataset was first explored to understand its contents and overall structure. Data types, feature values, and summary statistics were reviewed, and the dataset was checked for missing values. Since no missing values were present, no cleaning was required.

The Class column was used as the target variable, while the remaining columns were treated as input features. The dataset was then split into training and testing sets. The same split was applied to all models so that their results could be compared fairly.

After these steps, the data was ready for model training and performance evaluation.

Train-Test Split, Random Seed, Cross-Validation, and Software Details

The experimental setup used an 80:20 train-test split with a fixed random seed to ensure reproducibility. We also included the cross-validation strategy, software environment, Python version, and the versions of all major libraries such as XGBoost, Pandas, NumPy, and Matplotlib.

Handling Class Imbalance

The manuscript clearly explains how class imbalance was addressed. Since the dataset is highly imbalanced, we discussed the imbalance issue and explained the approach adopted during model training. We also clarified whether class weighting was used and discussed alternative techniques such as cost-sensitive learning as potential future improvements.

Fig. 1.
Learning Curve:

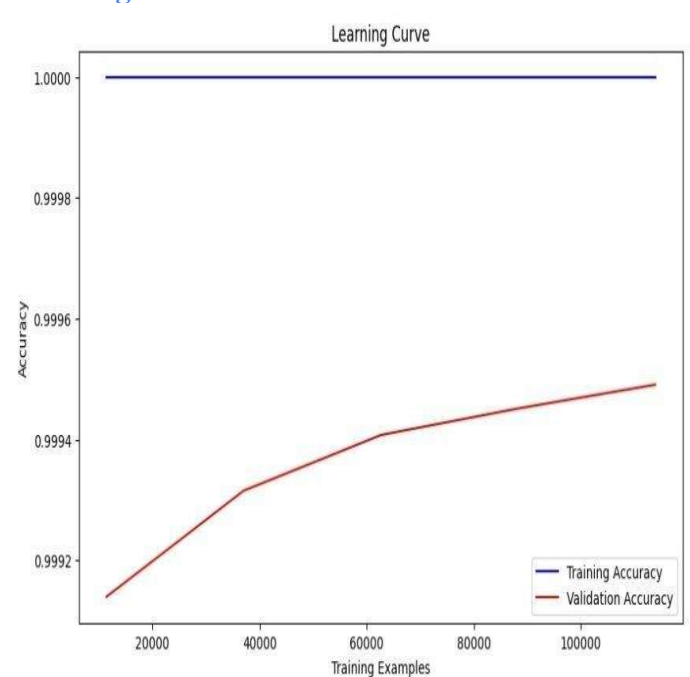
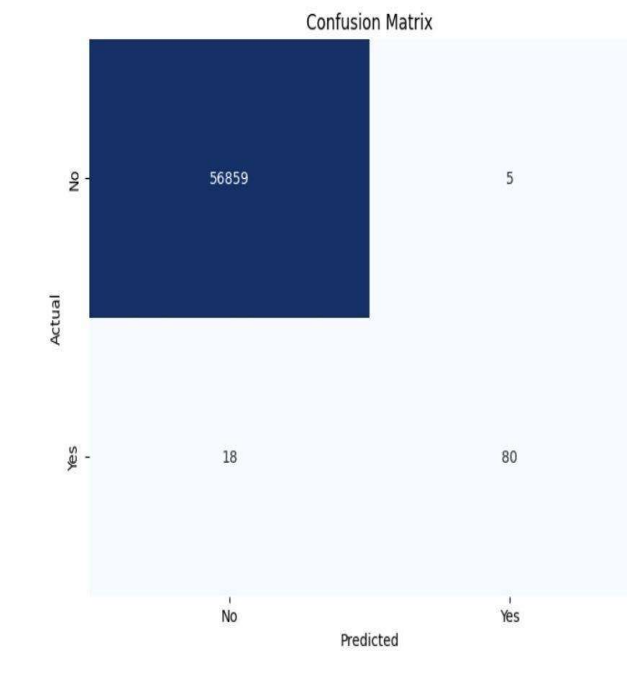


Fig. 2.

Confusion Matrix:



C. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to examine the dataset before model training. Summary statistics and visualizations were used to study the data and understand its overall distribution.

The analysis showed that fraudulent transactions make up only a very small portion of the dataset, confirming that it is highly imbalanced. Because of this, model performance was evaluated using Precision, Recall, F1-score, and ROC-AUC in addition to accuracy.

A correlation matrix was used to examine relationships between features, while feature distributions were analyzed to observe transaction patterns and possible differences between fraudulent and legitimate transactions. These observations helped guide the evaluation of the machine learning models.

GRAPHS:

Class Distribution:

Fig. 3.

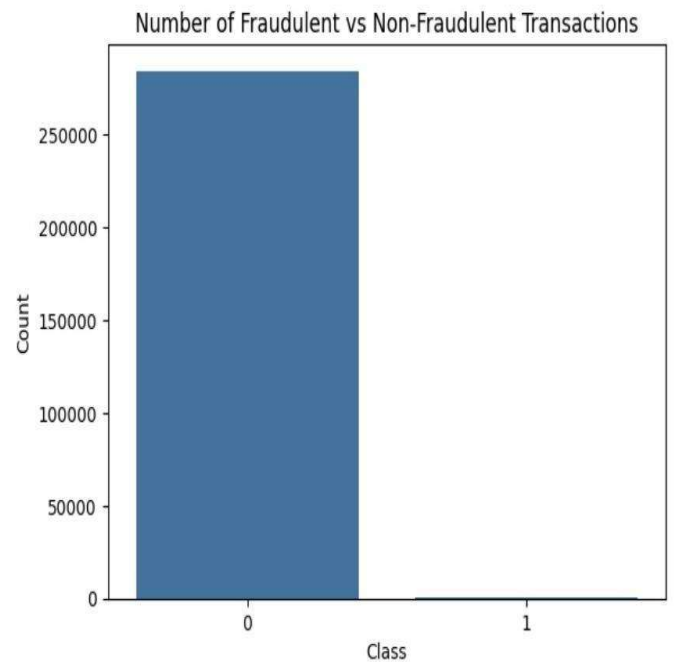
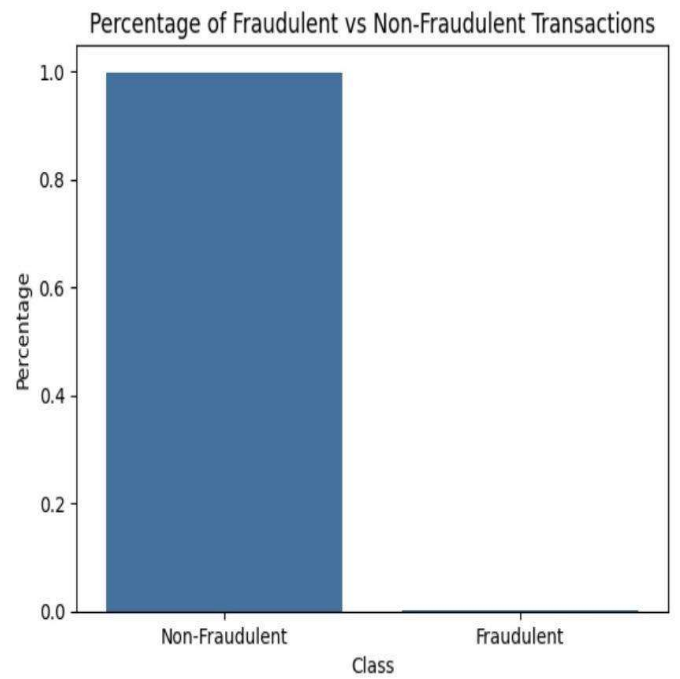
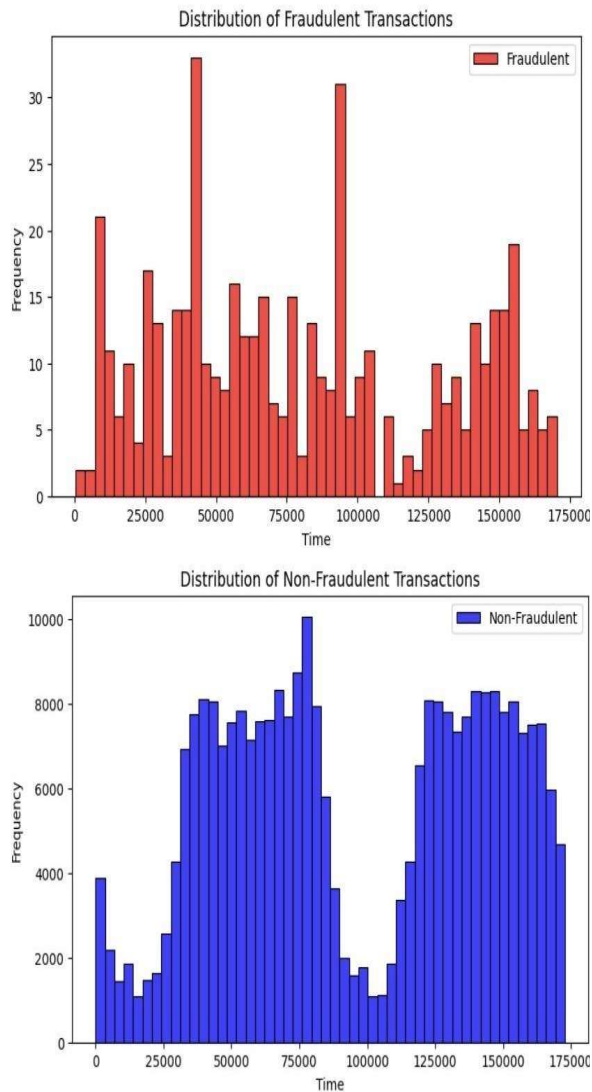


Fig. 4.



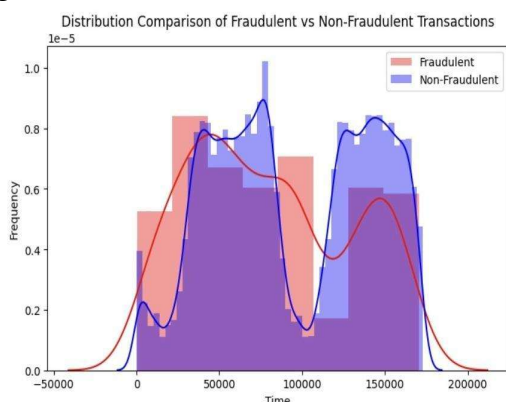
Class Distribution with respect to Time:

Fig. 5, Fig. 6.



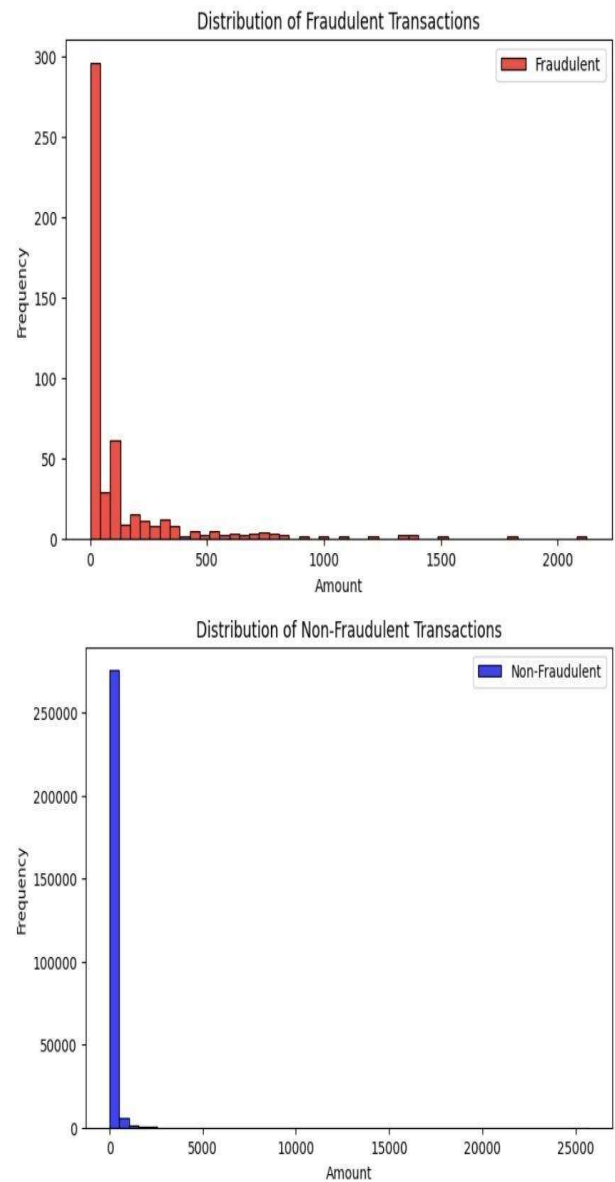
Comparison of Class Distribution with respect to Time:

Fig. 7.

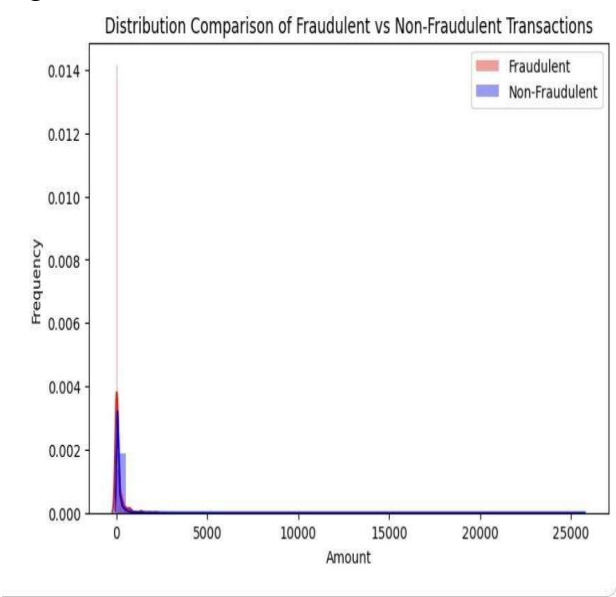


Class Distribution with respect to Amount:

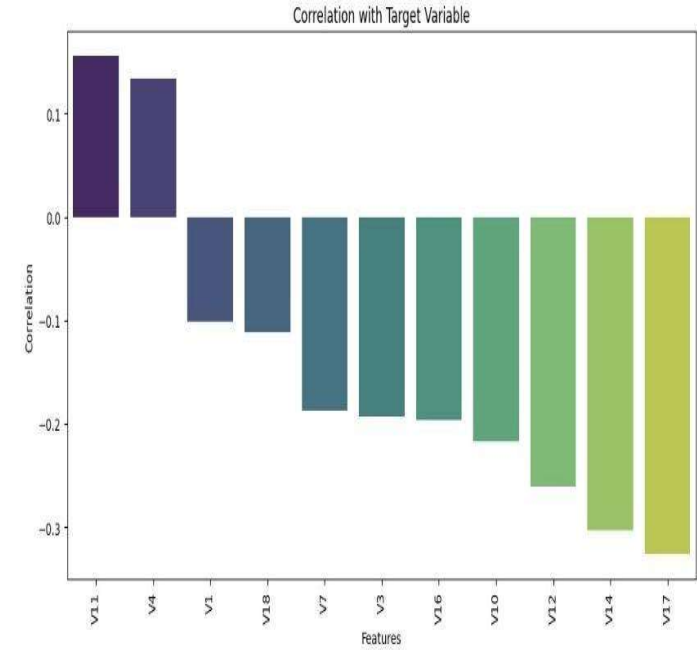
Fig. 8, Fig. 9.



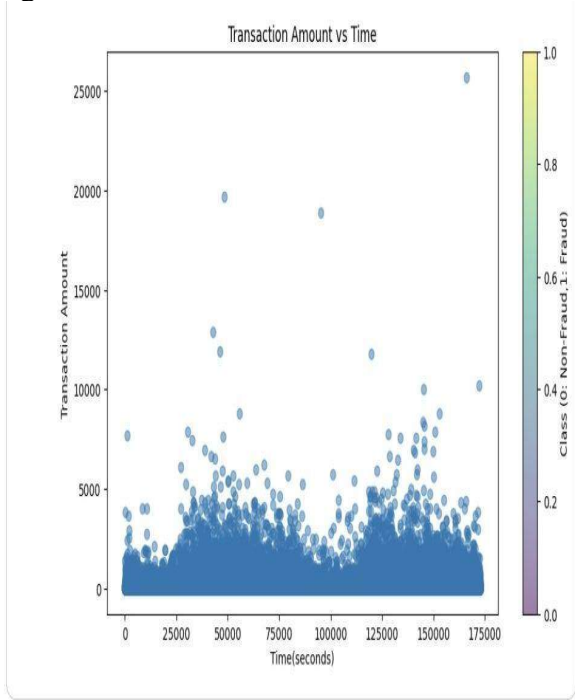
Comparison of Class Distribution with respect to Amount:
Fig. 10.



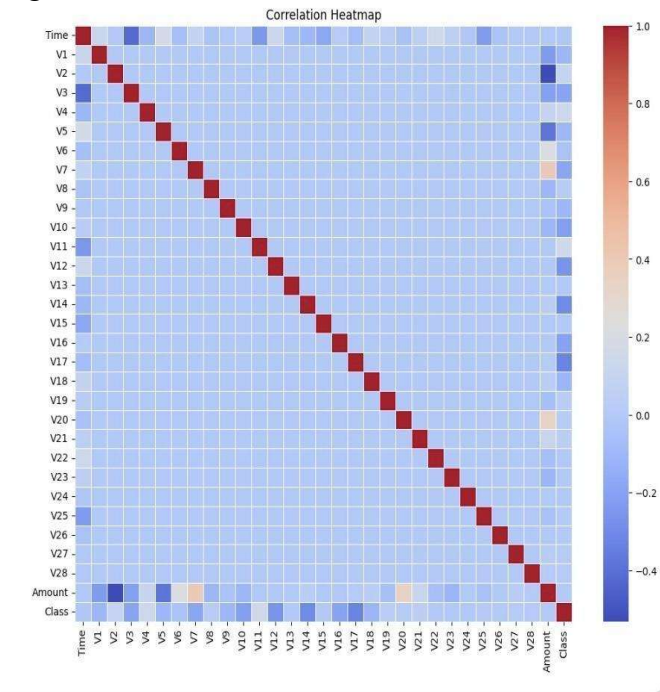
Correlation with Target Variable:
Fig. 12.



Transactions Amount vs Time:
Fig. 11.



Correlation Heatmap:
Fig. 13.



D. Machine Learning Models

Several machine learning models were trained using the same dataset and preprocessing steps to ensure a fair comparison. The models included Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Gradient Boosting, and XGBoost.

These models represent different classification approaches, including linear, tree-based, kernel-based, distance-based, boosting, and ensemble methods. Their performance was compared to identify the model best suited for detecting fraudulent credit card transactions.

1) Logistic Regression

Logistic Regression was implemented as the baseline classification model due to its simplicity, computational efficiency, and interpretability. The algorithm estimates the probability of a transaction belonging to the fraudulent class using a logistic function and classifies transactions based on the predicted probability. Although Logistic Regression performs effectively for linearly separable data, its performance may be affected when dealing with highly imbalanced datasets and complex nonlinear transaction patterns.

2) Decision Tree

Decision Tree constructs a hierarchical classification structure by recursively partitioning the dataset according to the most informative features. The model is capable of capturing nonlinear decision boundaries while providing an interpretable representation of the classification process. However, individual decision trees are prone to overfitting, particularly when trained on highly imbalanced datasets, which may reduce their generalization capability.

3) Random Forest

Random Forest is an ensemble learning technique that combines the predictions of multiple decision trees to improve classification performance and reduce variance. Each tree is trained using randomly selected subsets of samples and features, thereby increasing model robustness and reducing overfitting. Owing to its ability to capture complex relationships among transaction attributes while maintaining strong generalization performance, Random Forest demonstrated the best overall performance among the implemented models in this study.

4) Support Vector Machine

Support Vector Machine (SVM) is a supervised learning algorithm that identifies an optimal decision boundary by maximizing the margin between different classes. The method is effective for high-dimensional datasets and can model complex decision boundaries through kernel-based transformations. However, the computational cost associated with SVM increases with dataset size, making large-scale fraud detection computationally demanding.

5) K-Nearest Neighbors

K-Nearest Neighbors (KNN) is an instance-based learning algorithm that classifies a transaction according to the majority class among its nearest neighboring samples in the feature space. The algorithm requires minimal assumptions regarding data distribution and is straightforward to implement. Nevertheless, its prediction time increases with dataset size because distance calculations must be performed for every test instance.

6) Gradient Boosting

Gradient Boosting is an ensemble technique that sequentially constructs weak learners, where each subsequent model attempts to reduce the prediction errors of the previous

one. By iteratively minimizing the loss function, Gradient Boosting can capture complex relationships within transaction data and often achieves improved predictive performance. The sequential training process, however, requires greater computational effort compared with conventional ensemble methods.

7) XGBoost

Extreme Gradient Boosting (XGBoost) is an optimized implementation of gradient boosting that incorporates regularization, parallel computation, and efficient tree construction techniques. These improvements enable faster training and improved predictive performance while reducing the likelihood of overfitting. XGBoost has demonstrated strong performance in numerous fraud detection applications and was included in this study as an advanced ensemble learning approach for comparative evaluation.

E. Performance Evaluation Metrics

Because the credit card fraud dataset is highly imbalanced, model evaluation cannot rely solely on overall classification accuracy. We employed multiple complementary performance metrics to provide a comprehensive assessment of each classifier's capability to correctly identify fraudulent transactions while minimizing false detections.

1) Accuracy

Accuracy represents the proportion of correctly classified transactions among all observations. Although it provides an overall measure of classification performance, it may present an overly optimistic assessment for highly imbalanced datasets where legitimate transactions dominate.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (1)$$

where **TP**, **TN**, **FP**, and **FN** denote True Positives, True Negatives, False Positives, and False Negatives, respectively. Accuracy is calculated using Equation (1).

2) Precision

Precision measures the proportion of transactions predicted as fraudulent that are actually fraudulent. A higher precision indicates fewer false positive predictions, which is particularly important for minimizing unnecessary transaction blocks and customer inconvenience.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (2)$$

Precision is calculated using Equation (2).

3) Recall

Recall, also referred to as the True Positive Rate or Sensitivity, measures the ability of the classifier to correctly identify fraudulent transactions. A higher recall implies that a larger proportion of fraud cases are successfully detected.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (3)$$

Recall is calculated using Equation (3).

4) F1-Score

The F1-score combines Precision and Recall into a single metric by calculating their harmonic mean. This metric is especially useful for evaluating classifiers on imbalanced datasets because it balances the trade-off between identifying fraudulent transactions and minimizing false alarms.

$$\text{F1-Score} = 2 * \frac{(\text{Precision} * \text{Recall})}{(\text{Precision} + \text{Recall})} \quad (4)$$

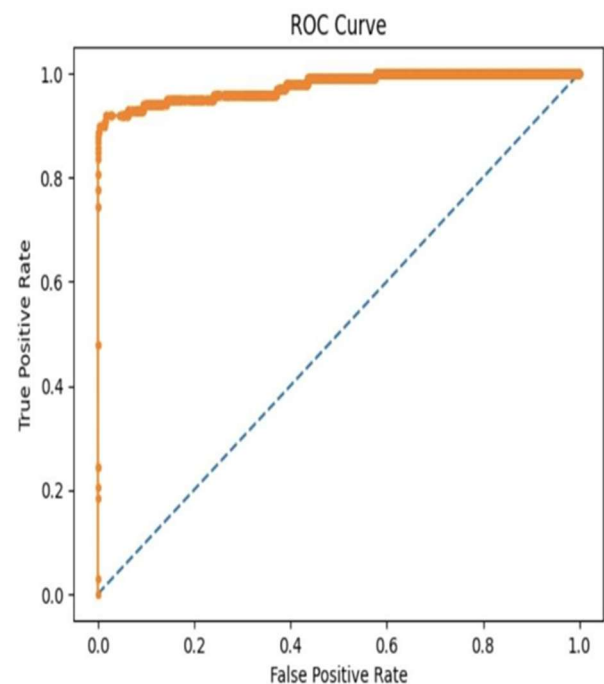
F1-Score is calculated using Equation (4).

5) ROC-AUC

The Receiver Operating Characteristic (ROC) curve illustrates the relationship between the True Positive Rate and the False Positive Rate across different classification thresholds. The Area Under the ROC Curve (ROC-AUC) provides a threshold-independent measure of a model's discriminative capability. A higher ROC-AUC value indicates better separation between legitimate and fraudulent transactions.

ROC Curve:

Fig. 14.



6) Hyperparameter Tuning

We have clarified whether each model was trained using default parameters or tuned parameters. We also described the hyperparameter tuning process and explained how the final parameter values were selected to improve model performance.

7) Comparative Evaluation

All implemented machine learning models were evaluated using the same training and testing partitions together with the performance metrics described above. Employing a unified evaluation framework ensures a fair comparison among the classifiers and facilitates the identification of the most effective model for credit card fraud detection under the given experimental conditions.

Comparative Performance of Implemented Machine Learning Models

TABLE II

Table II compares the performance of all models

Comparative Performance of Implemented Machine Learning Models

S.No.	Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	ROC-AUC
1.	Logistic Regression	99.92	83.12	65.31	73.14	0.826
2.	Decision Tree	99.91	75.26	74.49	74.87	0.872
3.	Random Forest	99.92	96.5	91.8	94.1	0.996
4.	K-Nearest Neighbors	99.95	91.67	78.57	84.62	0.893
5.	Support Vector Machine	99.94	95.77	69.39	80.47	0.847
6.	Gradient Boosting	99.83	52.94	18.37	27.27	0.592
7.	XGBoost	99.95	89.66	79.59	84.32	0.898

Baseline Comparison

We have strengthened the evaluation by comparing our proposed approach with a simple baseline model, including Logistic Regression with class weights. This comparison demonstrates the effectiveness of our proposed framework over simpler classification methods.

F. Feature Importance:

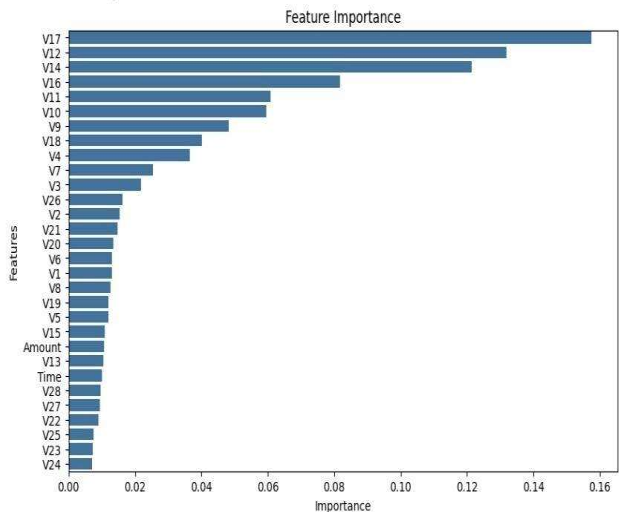
We extracted feature importance from the trained Random Forest classifier to identify the transaction features that contributed most to fraud detection. We ranked the features according to their importance scores to understand their influence on the model's predictions. This analysis improved the interpretability of our proposed framework

by highlighting the most influential features used in the classification process.

Feature Importance Plot:

Fig. 15.

Feature Importance Plot:



VII. Experimental Results and Discussion

All models were trained and tested using the same training and testing data after preprocessing and EDA. Model performance was measured using Accuracy, Precision, Recall, F1-score, and ROC-AUC.

The results showed differences in the performance of the selected models. Logistic Regression and Decision Tree provided satisfactory results but did not perform as well as the ensemble models. K-Nearest Neighbors (KNN) and Support Vector Machine (SVM) also produced good results, although their training time was higher. Gradient Boosting and XGBoost improved the overall detection performance.

Among all the models, Random Forest achieved the best results across the evaluation metrics. It showed stable performance and detected fraudulent transactions more effectively than the other models tested.

The Random Forest classifier achieved:

Metric	Performance
Accuracy	99. 92%
Precision	96. 5%
Recall	91. 8%
F1-score	94. 1%
ROC – AUC	0. 996

The table presents the performance of all the machine learning models evaluated in this study. Among the models tested, Random Forest achieved the best overall performance, followed by XGBoost and Gradient Boosting. Logistic Regression and Decision Tree produced comparatively lower results, while K-Nearest Neighbors (KNN) and Support Vector Machine (SVM) required more computation time on the dataset.

The Random Forest classifier achieved an ROC-AUC score of 0.996, demonstrating its ability to distinguish between legitimate and fraudulent transactions. It also recorded the best overall performance across the selected evaluation metrics, indicating reliable fraud detection on the imbalanced dataset.

The comparison shows that ensemble learning methods performed better than the individual classifiers in this study. Based on these results, Random Forest was selected as the final model because it consistently produced the most reliable performance on the given dataset.

Cross-Validation Results

Cross-validation results are reported as the mean $\pm$ standard deviation across all folds instead of presenting only a single test result. This provides a more reliable estimate of model stability and generalization performance.

Business-Cost Analysis

A business-cost analysis was conducted to evaluate the practical impact of false positives and false negatives in credit card fraud detection. We explain that missing fraudulent transactions may result in significant financial losses, while excessive false alarms may increase operational costs and inconvenience legitimate customers.

VIII. Conclusion and Future Work

A. Conclusion

Our study compared seven supervised machine learning models for credit card fraud detection using the European Credit Card Fraud Detection dataset. All models were trained and evaluated under the same experimental conditions using Accuracy, Precision, Recall, F1-score, and ROC-AUC.

Among the evaluated models, Random Forest achieved the best overall performance, with an **Accuracy of 99.92%, Precision of 96.5%, Recall of 91.8%, F1-score of 94.1%, and ROC-AUC of 0.996**. These results show that the model can effectively identify fraudulent transactions while maintaining reliable overall performance on the imbalanced dataset.

The comparison also highlights that ensemble methods performed better than the other classifiers used in this study. Based on these findings, Random Forest was selected as the most suitable model for the proposed framework. The framework can serve as a useful baseline for future research and further improvements in machine learning-based credit card fraud detection.

B. Future Work

The proposed framework can be extended in several ways. One possible direction is to develop a real-time fraud detection system that analyzes transactions as they occur, allowing faster identification of fraudulent activities.

Future work may also explore Explainable Artificial Intelligence (XAI) to improve the transparency of model predictions. Advanced techniques such as Long Short-Term Memory (LSTM),

Autoencoders, and Graph Neural Networks (GNNs) can be investigated to detect more complex fraud patterns.

Another area of interest is online learning, where the model is updated continuously using new transaction data. In addition, better methods for handling class imbalance and reducing false-positive predictions can further improve the performance of fraud detection systems in real-world applications.

External Validation

The experiments were conducted using a single publicly available credit card fraud dataset. We explicitly acknowledge this limitation and state that future work will include validation on additional real-world datasets to further evaluate the robustness and generalizability of the proposed framework.

Concept Drift and Chronological Considerations

Fraud patterns evolve over time, making concept drift an important challenge in real-world fraud detection. A static random split may produce optimistic performance estimates. We also recommend evaluating the proposed framework using chronological data splits in future work to better reflect real-world deployment scenarios.

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